WEEK 4 ASSIGNMENT

Name : Payal Vijay Rananaware

Title:

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

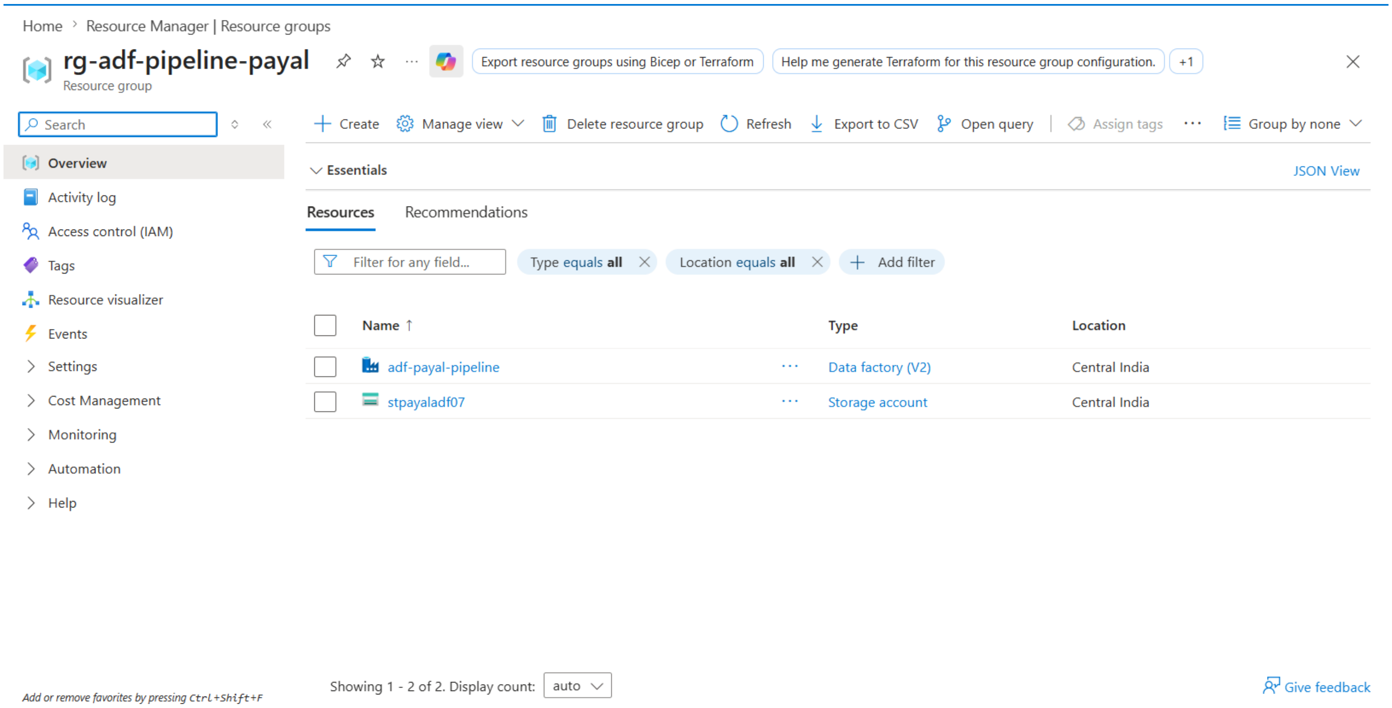
Assignment Tasks :-

Task 1:

- Explore Azure Portal
- Create a Resource Group

Deliverable:

- Screenshot of Resource Group



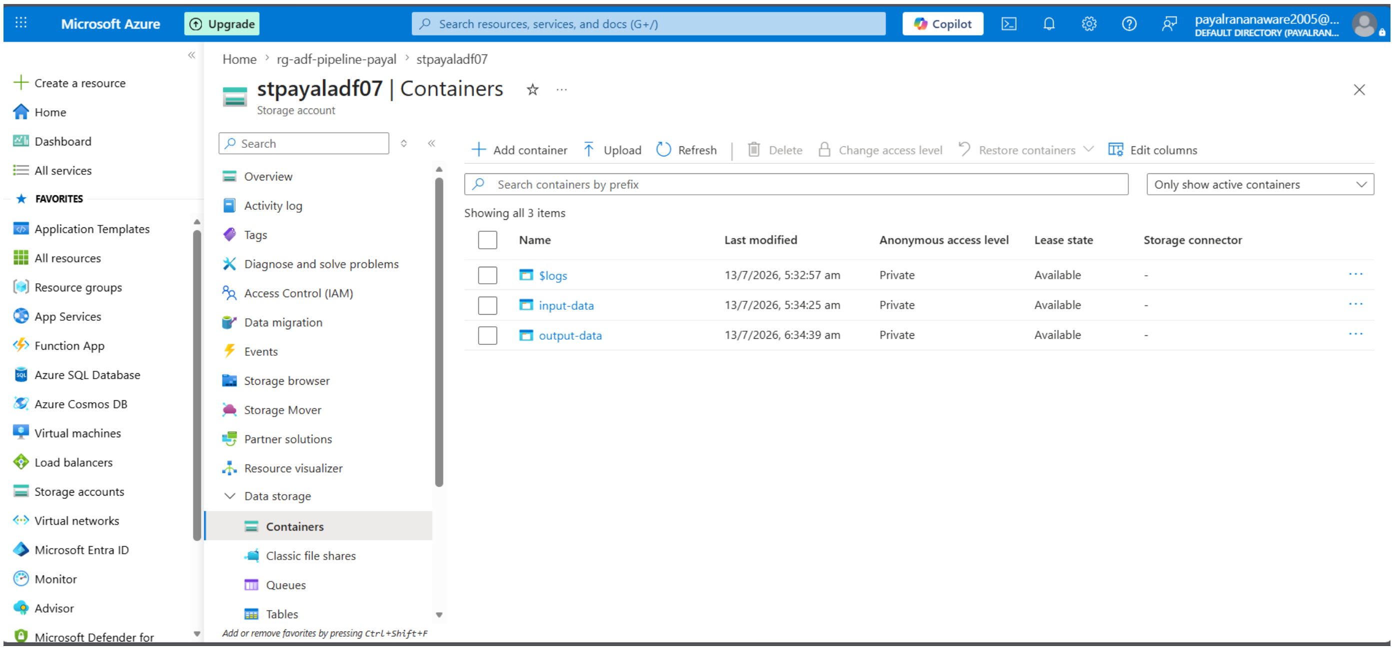
Task 2: Storage Setup

Do:

- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

Deliverable:

- Screenshot of container with uploaded file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Search resources, services, and docs (G+/)
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Create a resource
Home
Dashboard
All services
FAVORITES
Application Templates
All resources
Resource groups
App Services
Function App
Azure SQL Database
Azure Cosmos DB
Virtual machines
Load balancers
Storage accounts
Virtual networks
Microsoft Entra ID
Monitor
Advisor
Microsoft Defender for
Search
Overview
Activity log
Tags
Diagnose and solve problems
Access Control (IAM)
Data migration
Events
Storage browser
Storage Mover
Partner solutions
Resource visualizer
Data storage
Containers
Classic file shares
Queues
Tables
Add or remove favorites by pressing Ctrl+Shift+F
Add container
Upload
Refresh
Delete
Change access level
Restore containers
Edit columns
Search containers by prefix
Only show active containers
Showing all 3 items
Name
Last modified
Anonymous access level
Lease state
Storage connector
$logs
13/7/2026, 5:32:57 am
Private
Available
-
input-data
13/7/2026, 5:34:25 am
Private
Available
-
output-data
13/7/2026, 6:34:39 am
Private
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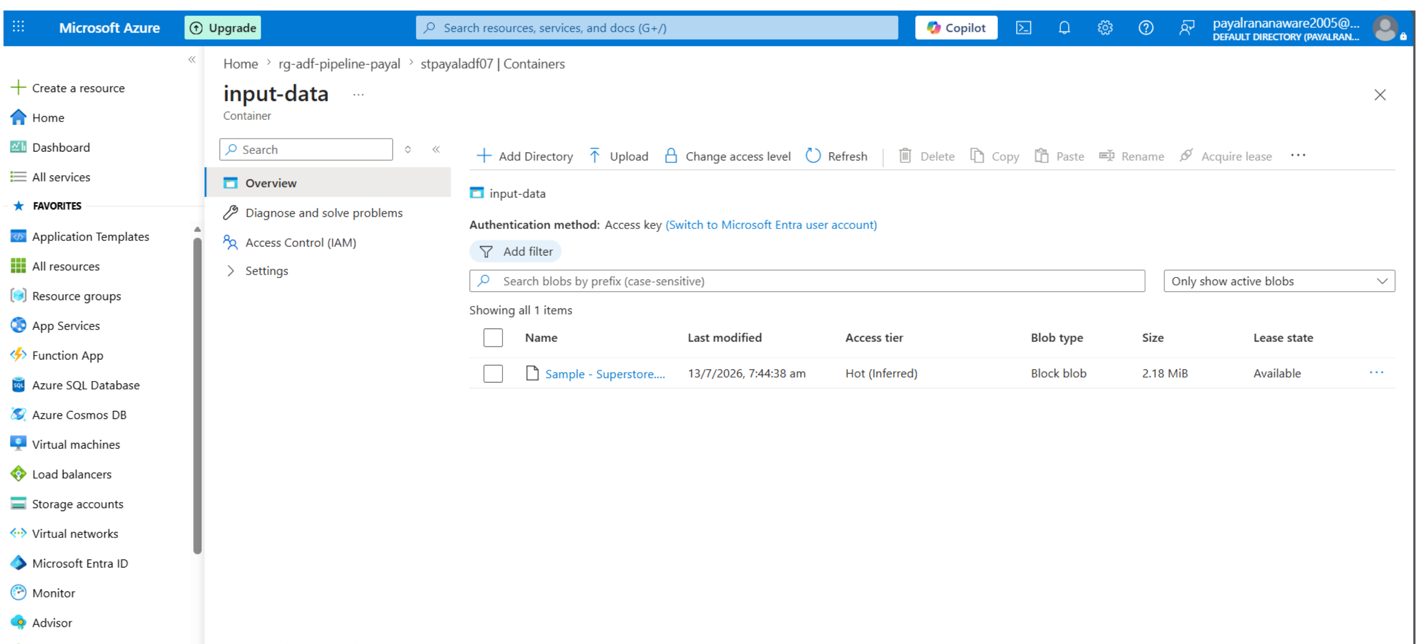
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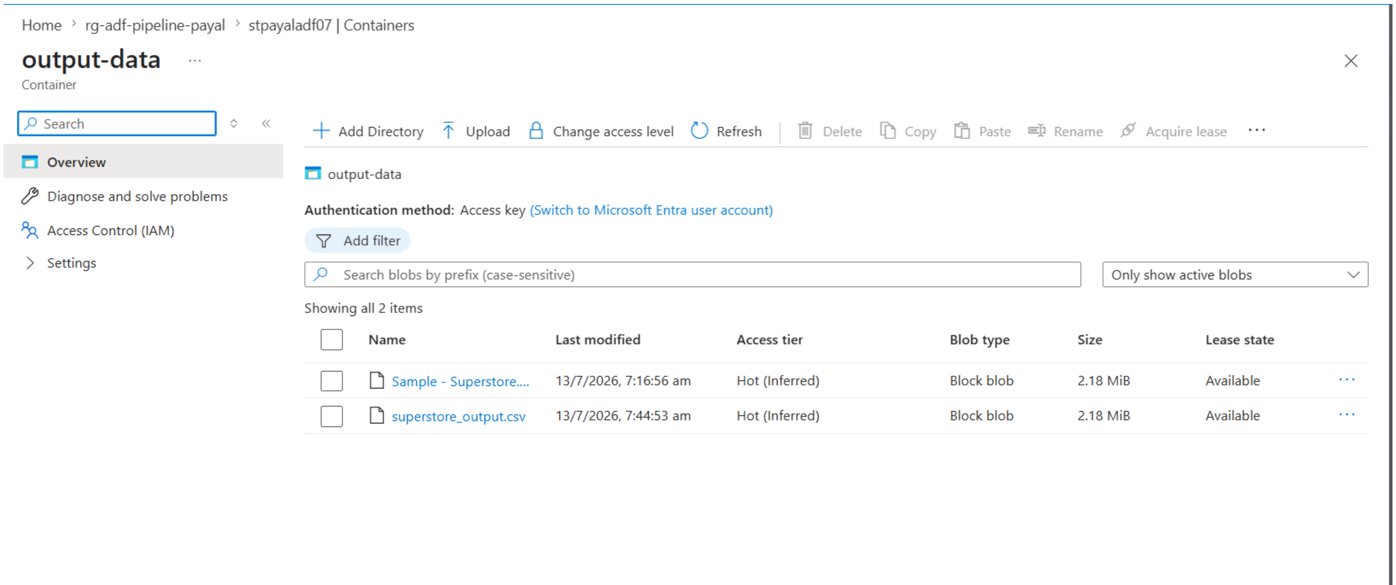
Storage accounts
Virtual networks
Microsoft Entra ID
Monitor
Advisor
Microsoft Defender for
Search
Overview
Diagnose and solve problems
Access Control (IAM)
Settings
Add or remove favorites by pressing Ctrl+Shift+F
Add Directory
Upload
Change access level
Refresh
Delete
Copy
Paste
Rename
Acquire lease
input-data
Authentication method: Access key (Switch to Microsoft Entra user account)
Add filter
Search blobs by prefix (case-sensitive)
Only show active blobs
Showing all 1 items
Name
Last modified
Access tier
Blob type
Size
Lease state
Sample - Superstore....
13/7/2026, 7:44:38 am
Hot (Inferred)
Block blob
2.18 MiB
Available



Task 3: ADF Basics

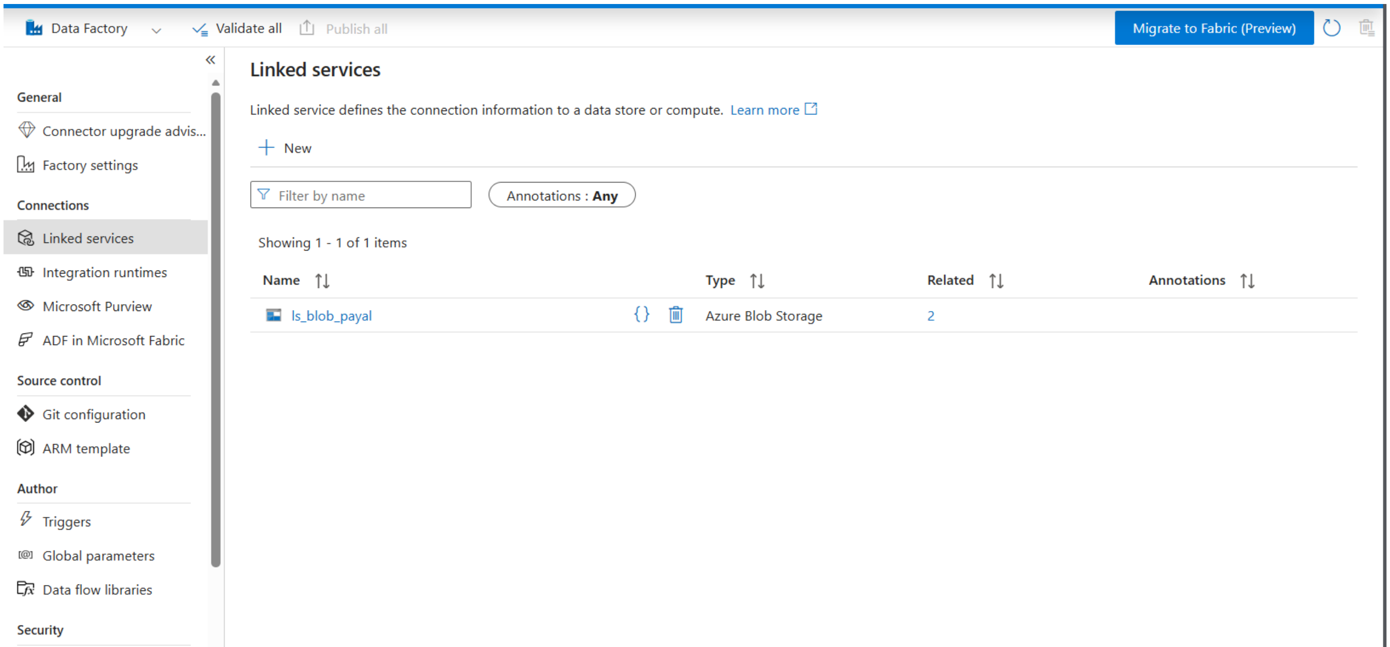
Do:

- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)
- Create datasets (source + destination)
- Use Get Metadata activity

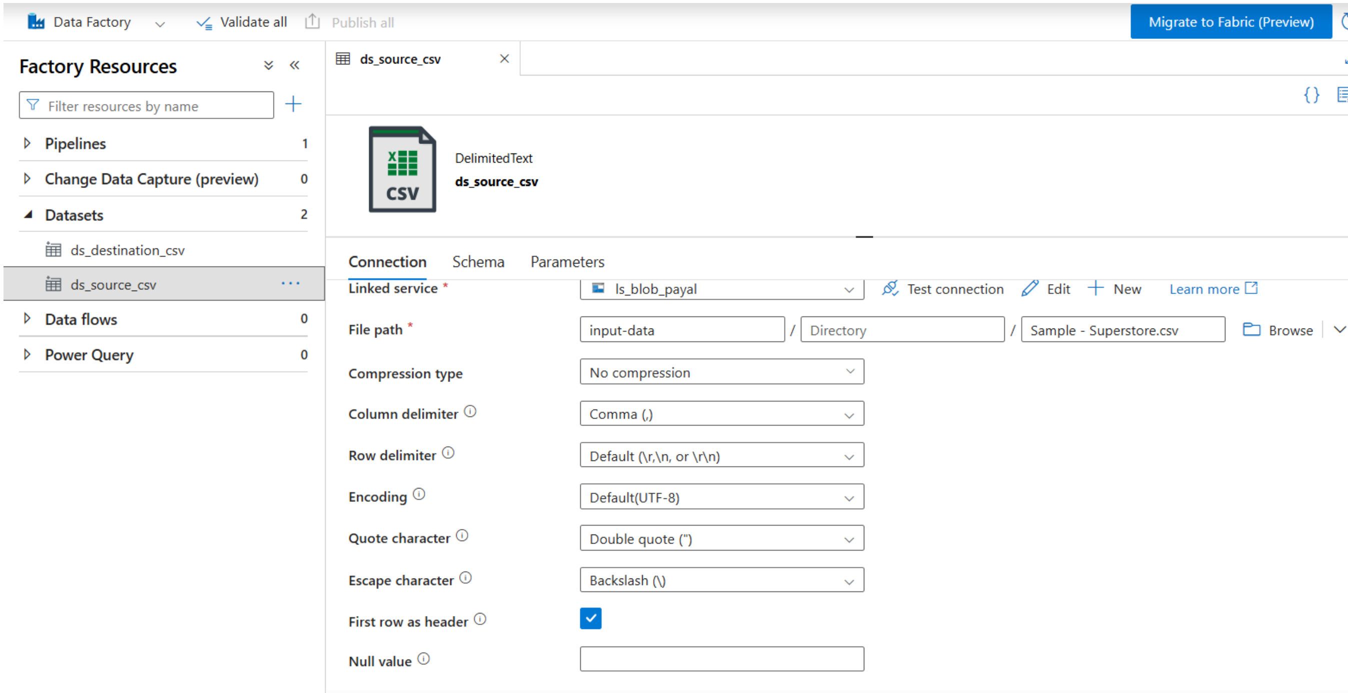
Deliverable:

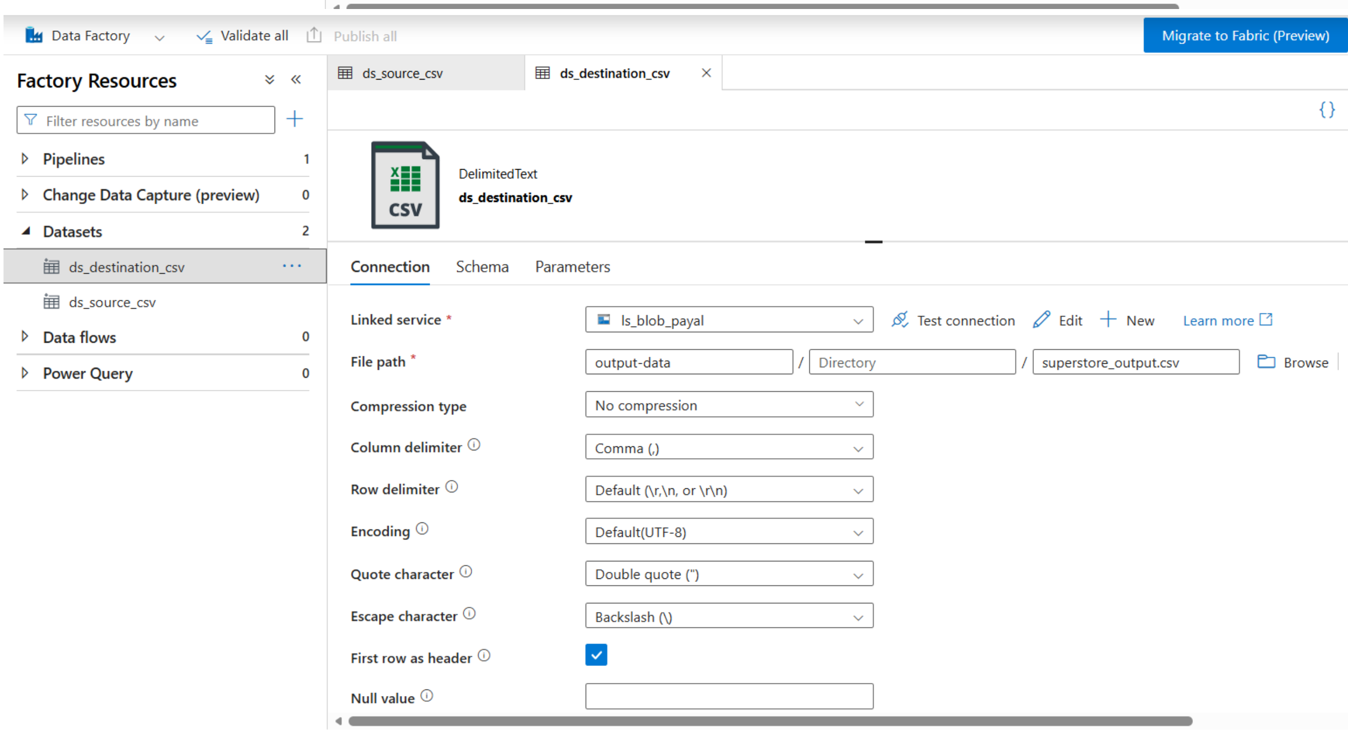
- Screenshot of Linked Service
- Screenshot of dataset
- Screenshot of Get Metadata activity

Screenshot of Linked Service :-

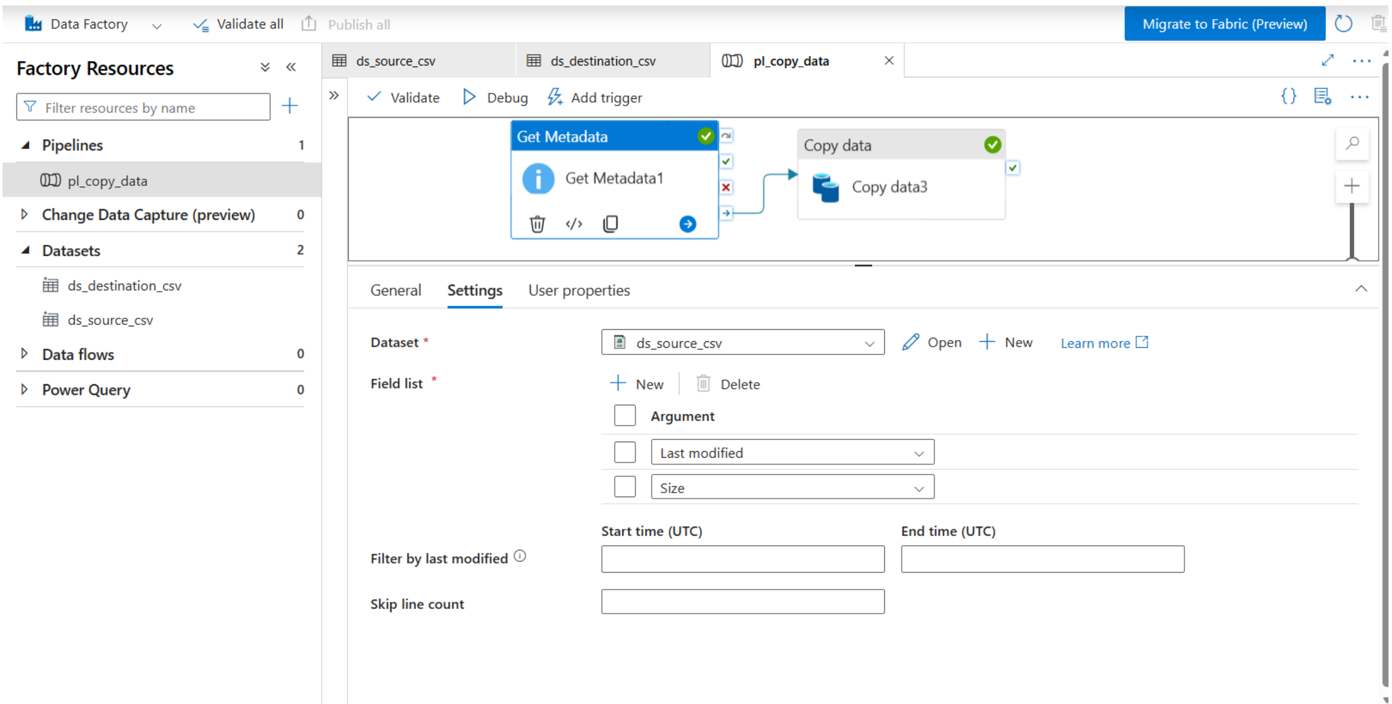


Screenshot of dataset :-

- 



Screenshot of Get Metadata activity :-



Output

Copy to clipboard


```
{
    "lastModified": "2026-07-13T03:22:36Z",
    "size": 2287806,
    "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (Central India)",
    "executionDuration": 0,
    "durationInQueue": {
        "integrationRuntimeQueue": 0
    },
    "billingReference": {
        "activityType": "PipelineActivity",
        "billableDuration": [
            {
                "meterType": "AzureIR",
                "duration": 0.016666666666666666,
                "unit": "Hours"
            }
        ]
    }
}
```

Task 4: Pipeline Development

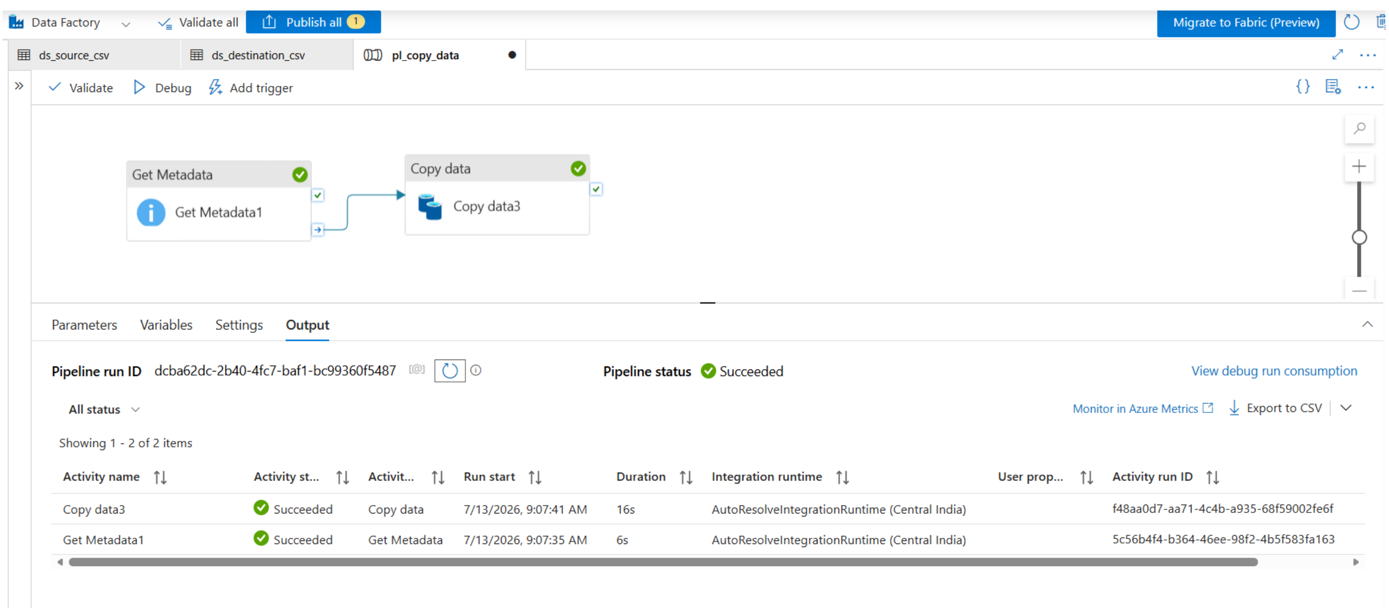
Do:

- Create pipeline using Copy Data activity
- Configure source and destination
- Add ForEach activity (optional if multiple files)

Deliverable:

- Screenshot of pipeline design

The ForEach activity was not used in the pipeline as the dataset contained only a single CSV file. The ForEach activity is typically used when multiple files need to be processed iteratively, which was not required in this case.



Output

Copy to clipboard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{
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    "size": 2287806,
    "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (Central India)",
    "executionDuration": 0,
    "durationInQueue": {
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    },
    "billingReference": {
        "activityType": "PipelineActivity",
        "billableDuration": [
            {
                "meterType": "AzureIR",
                "duration": 0.016666666666666666,
                "unit": "Hours"
            }
        ]
    }
}
```

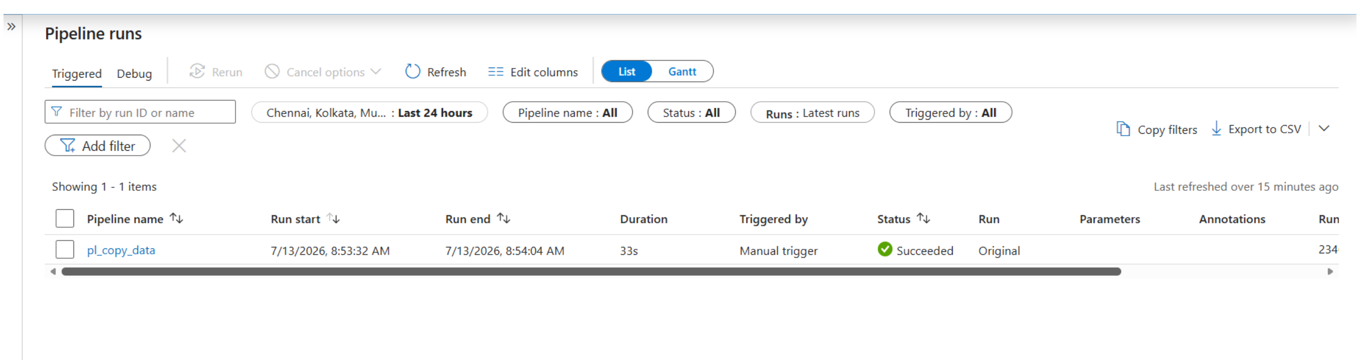

Task 5: Pipeline Execution

Do:

- Run pipeline using Debug/Trigger

Deliverable:

- Screenshot showing pipeline execution (Succeeded)



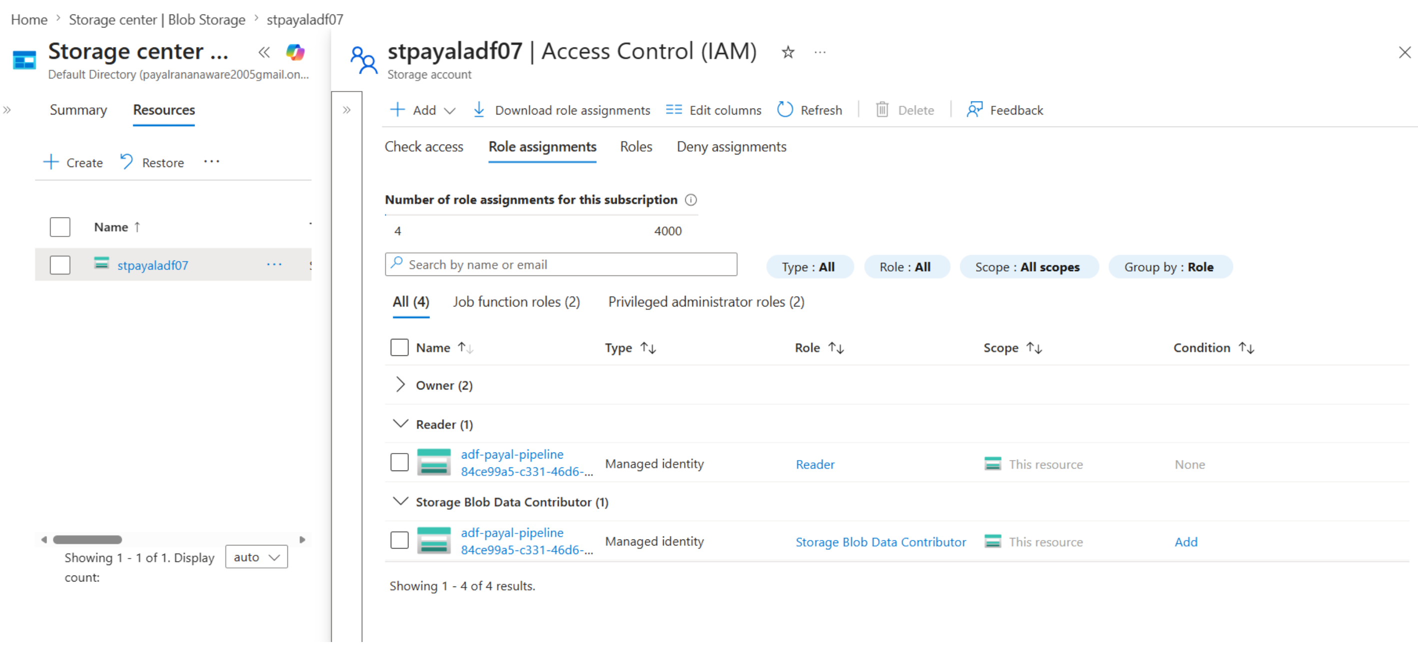
Task 6: IAM Roles

Do:

- Assign roles:
  - Reader
  - Contributor
- Provide access to ADF for Storage

Deliverable:

- Screenshot of role assignment



Mini Project (Final Task)

Problem Statement:

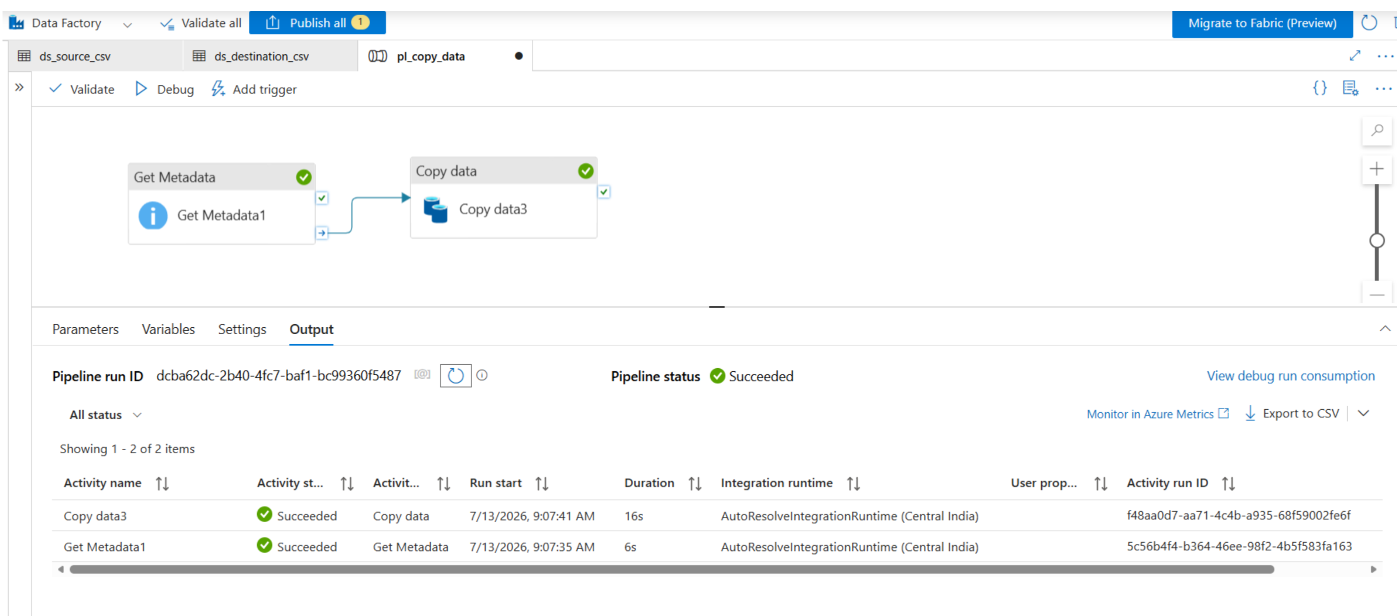
Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

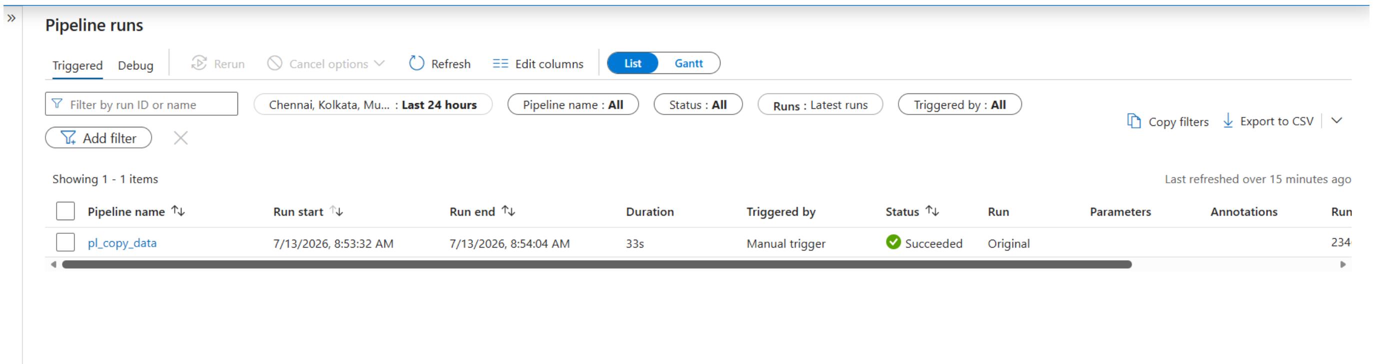
Requirements:

- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

Expected Output:

- Pipeline executed successfully
- Data copied to destination
- Metadata validated





Output

Copy to clipboard


```
{
    "lastModified": "2026-07-13T03:36:49Z",
    "size": 2287806,
    "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (Central India)",
    "executionDuration": 0,
    "durationInQueue": {
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    "billingReference": {
        "activityType": "PipelineActivity",
        "billableDuration": [
            {
                "meterType": "AzureIR",
                "duration": 0.016666666666666666,
                "unit": "Hours"
            }
        ]
    }
}
```

Summary

- Implemented an end-to-end Azure Data Factory pipeline using Blob Storage.
- Created Resource Group, Storage Account, Blob containers, Linked Service, and datasets.
- Used Get Metadata for file validation and Copy Data for transferring CSV data from source to destination.
- Successfully executed the pipeline and verified the output file.
- Configured IAM roles to provide required access between Azure Data Factory and Storage.

Conclusion

The Azure Cloud Fundamentals and Data Pipeline Implementation task was successfully completed using Azure Storage Account and Azure Data Factory. An end-to-end data pipeline was created to transfer CSV data from Blob Storage to a destination location using Copy Data activity, with Get Metadata activity used for file validation. The pipeline was executed successfully, and the required access permissions were configured using IAM roles. Through this implementation, I gained practical understanding of Azure cloud resources, data pipeline development, monitoring, and secure data access management.